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SYSTEM FOR GENERATING RANDOM NUMBERS BASED ON BLOOD SAMPLES AND METHOD USING THE SAME

Abstract

According to an aspect of the present invention, there is provided a system for generating random numbers based on blood samples including a sample loading module configured to supply a preset blood sample, a light irradiation module configured to irradiate the blood sample provided by the sample loading module with light, the blood sample moved to a pre-designated region of interest (ROI) being irradiated with light, an image generation module configured to acquire an original speckle image by capturing an image of the blood sample located in the region of interest in a preset method using the light irradiation module, and a random number generation module configured to generate random numbers by processing the original speckle image generated from the image generation module in a preset method.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims priority to Korean Patent Applications No. 10-2024-0030450, filed on Mar. 4, 2024, and 10-2024-0050017, filed on Apr. 15, 2024, all the benefits accruing therefrom under 35 U.S.C. § 119, the contents of which are incorporated by reference in their entirety.

BACKGROUND

[0002] The present disclosure relates to a system for generating random numbers based on blood samples and a method using the same, and more particularly, to a technology related to a random number generation system using a two-pass tuple-output von Neumann (2P-TO-VN) method.

[0003] The content described in this background section simply provides background information for the embodiments and does not constitute prior art.

[0004] As the use of wired and wireless communications, including the Internet, is rapidly expanding, the importance of communication network security issues is increasing in terms of protecting important national, corporate, and financial secrets and privacy of individuals. A cryptographic system based on mathematical computational complexity, such as RSA, is considered to be fundamentally decipherable once a quantum computer is developed, and thus countermeasures are required.

[0005] Compared to the cryptographic system based on computational complexity, a cryptograph method using one time pad (OTP) is known to be the most secure cryptographic method. When a stream cipher is generated using the one time pad, encrypted communication can be performed quickly and securely. A random number generator (RNG) is used to generate this one time pad.

[0006] The evolution and continued development of random number generator systems is laying an important cornerstone for the future of cryptography. This RNG system forms the backbone of all encryption systems due to its role in generating keys that are not only difficult to predict, but also nearly impossible to predict.

[0007] Such unpredictability can ensure a high level of security, which is a necessary feature in today's digital world. There are mainly two types of random number generators of algorithmic random number generators and physical random number generators. Algorithmic random number generators also known as pseudorandom number generators (PRNG) use complex mathematical formulas to generate a series of numbers.

[0008] The PRNG is convenient and efficient, but it has the disadvantage that the generated random numbers can be reverse engineered and predicted even if they pass a randomness test. On the other hand, the physical (or real) random number generators can derive randomness from stochastic physical processes. Although these processes are theoretically predictable with complete information, they are practically unpredictable due to limitations in time and computational resources.

[0009] Due to these characteristics, RNGs play an important role in a variety of cybersecurity tasks, including key generation, digital signature generation, initialization vectors for encryption, and salt value generation for secure storage, and can address the cybersecurity requirements for interconnected systems beyond the IoT domain.

[0010] One emerging area of interest is the use of optical and physical unclonable function to enhance security (Di Falco et al). It is illustrated that a chaotic system facilitated by a silicon chip can be exploited to achieve an encryption system having perfect secrecy.

[0011] However, it is important to note that physics-based RNG has its own unique challenges. For example, capturing random patterns in a lava lamp is an innovative method, but bandwidth is

limited. Another method of using light-emitting diodes and a cell phone camera, captures the randomness in quantum fluctuations of light, but the setup is incredibly complicated. Therefore, the environment of RNG systems has been changing in recent years, and existing artificially created RNGs have been found to contain unique security vulnerabilities.

[0012] Because of these vulnerabilities, the transition to a more unpredictable and inherently random bio-inspired RNG systems has been triggered. Changes acknowledge that a biological system often contains a level of complexity and randomness that, if not impossible, are difficult to artificially replicate.

[0013] Therefore, recently, the RNG technology is attempting to focus on generating RNGs using speckle patterns. The main goal of these technologies is to achieve a balance between affordability and portability. Both are considered essential for widespread adoption and application of these systems.

[0014] Advances in RNGs and physical de-cloning function of photons in the area of speckle pattern generation often rely on the inherent randomness of materials, such as the natural texture of paper, as speckle patterns are generated under coherent light. For example, optical waveguides demonstrate the ability to generate random numbers at Mbit/s with verified randomness.

[0015] In addition, an all-optical physical de-cloning function (Fratalocchi et al.) was developed based on the speckle pattern of aerogel to achieve secure key generation. However, these technologies involve passing a laser beam through a volumetric scattering medium to capture the static speckle pattern. Although this approach is innovative, it may pose new challenges for cost-effectiveness and high-speed random number generation. Furthermore, despite the improved security functions provided by these RNGs, their application in industrial settings is limited. The main obstacle lies in the high costs associated with sample production and acquisition speed. If these costs increase, purchasing speed generally improves.

[0016] Conversely, lower costs often result in slower speeds, which creates a trade-off that complicates optimization. This problem requires an RNG system that provides excellent performance while significantly reducing manufacturing costs.

[0017] As a prior art in the field of RNG, Korean registered patent No. 10-1925787 (Method and device for securing continuity of random number output signal during von Neumann post-processing) is disclosed. The prior art discloses solving the discontinuity of the random number output signal, and has an effect of improving the speed of the system and economizing to lower the cost by storing random numbers in a memory buffer and outputting spare random numbers through another configuration when all the stored random numbers are exhausted,

[0018] However, the prior art only discloses random number output technology using von Neumann post-processing, and does not disclose at all the possibility of applying and combining blood flow or blood samples.

SUMMARY

[0019] The present disclosure provides a system that enables high-speed random number generation while maximizing cost-efficiency and also ensures the reliability of the generated random numbers by improving the randomness of the generated random numbers.

[0020] However, aspects of the present invention are not limited to the system described above, and other aspects not mentioned will be apparent to those skilled in the art from the following description of the invention.

[0021] In accordance with an exemplary embodiment of the present invention, a system for generating random numbers based on blood samples includes a sample loading module configured to supply a preset blood sample, a light irradiation module configured to irradiate the blood sample provided by the sample loading module with light, the blood sample moved to a pre-designated region of interest (ROI) being irradiated with light, an image generation module configured to acquire an original speckle image by capturing an image of the blood sample located in the region of interest in a preset method using the light irradiation module, and a random number generation

module configured to generate random numbers by processing the original speckle image generated from the image generation module in a preset method.

[0022] Preferably, the random number generation module may include a binary image generation unit configured to generate a binary image by performing a binarization process of classifying pixel values constituting the original speckle image into bit of 0 or 1.

[0023] Preferably, the random number generation module may further include a von Neumann extraction unit configured to generate a first processed image by applying a von Neumann post-processing method to the binary image acquired through the binary image generation unit.

[0024] Preferably, the von Neumann extraction unit may be configured to generate a second processed image by further applying a two-pass tuple-output von Neumann (2P-TO-VN) method to the first processed image.

[0025] Preferably, the von Neumann extraction unit may be configured to apply a one-pass tuple-output von Neumann (1P-TO-VN) method and process two bits as one set in a preset method, and the preset method may be a method of removing one set if the one set is a duplication of the same bits.

[0026] Preferably, the von Neumann extraction unit may be configured to, when a set is composed of different bits based on the remaining bits in a state where one set is the duplication of the same bits and the set is removed, maintain only a first bit or second bit constituting the set among the first bit and second bit.

[0027] Preferably, the von Neumann extraction unit may be configured to regroup the removed set into a quad through the 1P-TO-VN method, and the quad is composed of a first set and a second set, and determine whether to maintain the removal of the first and second sets based on whether the first and second sets are the same.

[0028] Preferably, the von Neumann extraction unit may be configured to restore the first and second sets when the first and second sets are different from each other.

[0029] Preferably, the random number generation module may be configured to further include an image scrambling unit configured to generate a pre-processed image by performing image scrambling on the binary image generated through the binary image generation unit, and the von Neumann extraction unit may be configured to generate the first and second processed images for the pre-processed image generated by the image scrambling unit.

[0030] In accordance with another exemplary embodiment of the present invention, a method of generating random numbers based on blood samples includes (a1) supplying a blood samples through a sample loading module, (a2) irradiating the blood sample provided by the sample loading module with light through a light irradiation module, the blood sample moved to a pre-designated region of interest (ROI) being irradiated with light, (a3) acquiring an original speckle image by capturing an image of the blood sample located in the region of interest in a preset method using the light irradiation module, and (a4) generating random numbers by processing the original speckle image generated in the (a3) in a preset method.

[0031] Preferably, the (a4) may include (a41) generating a binary image by performing a binarization process of classifying pixel values constituting the original speckle image into bit of 0 or 1, (a42) generating a first processed image by applying a von Neumann post-processing method to the binary image acquired in the (a41), and (a43) generating a second processed image by further applying a 2P-TO-VN scheme to the first processed image generated in the (a42).

[0032] Preferably, the method may further include, before the (a42), (a42-0) generating a pre-processed image by performing image scrambling on the binary image generated in the (a41).

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0033] Exemplary embodiments can be understood in more detail from the following description taken in conjunction with the accompanying drawings, and the following drawings attached to this specification illustrate preferred embodiments of the present invention, and serve to further understand the technical idea of the present invention together with the detailed description of the invention described later. Therefore, the present invention should not be construed as limited to the matters described in such drawings, in which:

[0034] FIG. 1 is a block diagram illustrating the overall configuration of the system in accordance with an exemplary embodiment of the present invention;

[0035] FIG. 2 is a schematic diagram illustrating a blood sample testing device of the system in accordance with the exemplary embodiment of the present invention;

[0036] FIG. 3 is a schematic diagram illustrating a sample loading module in the blood sample testing device of FIG. 2;

[0037] FIG. 4 is a conceptual diagram schematically illustrating a step of generating random numbers using the system in accordance with the exemplary embodiment;

[0038] FIG. 5 is a block diagram schematically illustrating a processing process in a von Neumann extraction unit of the system in accordance with the exemplary embodiment; FIG. 6 is an example for describing the process in FIG. 5;

[0039] FIG. 7 is a flowchart of a method in accordance with another exemplary embodiment of the present invention;

[0040] FIG. 8 is a flowchart dividing step S40 of FIG. 7 in detail;

[0041] FIG. 9A, FIG. 9B, FIG. 9C and FIG. 9D are a graph obtained by comparing the speckle decorrelation times of an original speckle image (OSI), a first processed image (CVN method), and a second processed image (2P-TO-VN method);

[0042] FIG. 10A, FIG. 10B, FIG. 10C and FIG. 10D are a graph obtained by comparing random number matrix characteristics of the original speckle image (OSI), the first processed image (CVN method), and the second processed image (2P-TO-VN method);

[0043] FIG. 11A, FIG. 11B, FIG. 11C, FIG. 11D and FIG. 11E illustrate results of speckle bit generation rates of the first processed image (CVN method) and the second processed image (2P-TO-VN method) and the respective images;

[0044] FIG. 12A is an exemplary image illustrating the original speckle image (OSI), the first processed image (CVN method), and the second processed image (2P-TO-VN method), and

[0045] FIG. 12B illustrates an exemplary image to which image scrambling has been applied; and

[0046] FIG. 13A and FIG. 13B are a table illustrating the results of the NIST statistical randomness test for random numbers generated using the system in accordance the exemplary embodiment of the present invention.

DETAILED DESCRIPTION OF EMBODIMENTS

[0047] Hereinafter, preferred embodiments of the present invention will be described in detail with reference to the attached drawings. Prior to this, terms and words used in this specification and claims should not be construed as limited to their usual or dictionary meanings, and should be interpreted as meanings and concepts consistent with the technical idea of the present invention based on the principle that the inventor can appropriately define terminology in order to explain his or her invention in the best way.

[0048] The present invention may be subjected to various modifications and have several embodiments, and specific embodiments thereof will be illustrated in the drawings and described in detail in the detailed description. However, this is not intended to limit the present invention to specific embodiments, and should be understood to include all modifications, equivalents, and substitutes that fall within the spirit and technical scope of the present invention. In describing each drawing, similar reference numerals are used for similar components.

[0049] Terms such as first, second, A, B, etc. can be used to describe various components, but the components should not be limited by the terms. The terms are used only to distinguish one component from another. For example, a first component may be named a second component without departing from the scope of rights of the present invention, and similarly, a second component may also be named a first component. The term “and/or” includes a combination of a plurality of related stated items or any of the plurality of related stated items.

[0050] When a component is referred to as being “coupled” or “connected” to another component, it should be understood that the component may be directly coupled or connected to that other component, but other components may exist in between. On the other hand, when a component is referred to as being “directly coupled” or “directly connected” to another component, it should be understood that no other components exist in between.

[0051] The terms used in this application are only used to describe specific embodiments and are not intended to limit the present invention. Singular expressions include plural expressions unless the context clearly dictates otherwise. It should be understood that, in this application, terms such as “comprise” or “have” are intended to designate the presence of features, numbers, steps, operations, components, parts, or combinations thereof described in the specification, and do not exclude in advance the possibility of the existence or addition of one or more other features or numbers, steps, operations, components, parts, or combinations thereof.

[0052] Unless otherwise defined, all terms used herein, including technical or scientific terms, have the same meaning as generally understood by a person of ordinary skill in the technical field to which the present invention pertains. Terms such as those defined in commonly used dictionaries should be interpreted as having a meaning consistent with the meaning in the context of the related technology, and unless clearly defined in this application, should not be interpreted in an ideal or excessively formal sense.

[0053] FIG. 1 is a block diagram illustrating the overall configuration of the system according to an embodiment of the present invention.

[0054] A system according to an embodiment of the present invention will be described with reference to FIG. 1. A system **10** includes a sample loading module **400**, a light irradiation module **100**, an image generation module **200**, and a random number generation module **300**. Here, the random number generation module **300** may be executed as a program stored in a processor included in the system.

[0055] FIG. 2 is a schematic diagram illustrating a blood sample testing device of the system according to embodiment of the present invention, and FIG. 3 is a schematic diagram illustrating a sample loading module in the blood sample testing device of FIG. 2.

[0056] Description will be made with reference to FIGS. 2 and 3 together.

[0057] The light irradiation module **100** may be configured to irradiate a blood sample S with light. As an example, the blood in the blood sample S may exemplarily include red blood cells R, platelets P, etc., as illustrated in FIG. 3. In addition, the light may be laser light having a wavelength within a certain range (e.g., 532 nm), but is not limited thereto.

[0058] The blood sample S may be extracted from a rat and contained in distilled deionized water, but is not limited thereto. The Applicant states that protocol GIST-2019-015 was strictly followed and that all officially approved methods were complied with ARRIVE guidelines for reporting experiments.

[0059] Specifically, the Applicant collected blood samples from approximately 12 to 13-week-old male Sprague Dawley rats weighing between approximately 250 and approximately 280 g via tail vein phlebotomy. For this purpose, 1 ml of blood was collected through a 23G needle while the animals were anesthetized with isoflurane. This blood collection was performed on a group of approximately 11 mice, and samples were immediately stored in citrate tubes (Catalog #363083, 9NC 0.109M Buffered Trisodium Citrate, BD Vacutainer, USA) for subsequent experiments.

[0060] The image generation module **200** may be configured to acquire a speckle image of the

blood sample S irradiated with light. Specifically, the image generation module **200** may include a photographing unit **210**, a first lens **220**, an aperture **230**, a second lens **240**, a polarizing plate **250**, a reflecting mirror **260**, a collimator **270**, and a coupler **280**.

[0061] The photographing unit **210** may be configured to photograph and acquire a speckle image of the blood sample S irradiated with light. As an example, the photographing unit **210** may be a CMOS camera (Neo 5.5 sCMOS, Andor Technology Ltd., Belfast, UK), but is not limited thereto. This CMOS camera operates at a high frame rate of approximately 1250 frames per second with an exposure time of approximately 0.8 milliseconds to smoothly capture speckle images with a resolution of approximately 128×512 pixels for subsequent data generation.

[0062] The speckle image of the blood sample S may include a speckle pattern that is irregularly generated by interference that occurs when light is reflected from the blood sample S or light passes through the blood sample S. As an example, a total of approximately 7 experimental data sets were constructed with approximately 1000 frames, and for pre-processing the experimental data, they were divided into approximately 8 partitions, each consisting of approximately 32×256 pixels, for a total of approximately 56 sets were used.

[0063] The first lens **220** is an objective lens and may be disposed between the photographing unit **210** and the sample loading module **400** on which the blood sample S is disposed. The aperture **230** is a diaphragm and may be disposed between the photographing unit **210** and the first lens **220**. The second lens **240** is a tube lens and may be disposed between the photographing unit **210** and the aperture **230**.

[0064] The polarizing plate **250** may be disposed between the second lens **240** and the aperture **230**.

[0065] In this case, the aperture **230**, the second lens **240**, and the polarizing plate **250** may be provided to improve contrast in the speckle image of the blood sample S acquired by the photographing unit **210**.

[0066] The reflecting mirror **260** may be configured to reflect light emitted from the light irradiation module **100** and make the light incident on the blood sample S. In this case, at least one reflecting mirror **260** may be provided.

[0067] The collimator **270** may be disposed between the sample loading module **400** on which the blood sample S is disposed and the reflecting mirror **260**.

[0068] The coupler **280** is an optical coupler and may be disposed between the collimator **270** and the reflecting mirror **260**.

[0069] The sample loading module **400** may be configured to move the blood sample S. As an example, the sample loading module **400** may contain a polydimethylsiloxane (PDMS) material. The sample loading module **400** may include a body **410**, an injection unit **420**, a movement channel **430**, and a discharge unit **440**.

[0070] The body **410** may configure the overall shape of the sample loading module **400**. The injection unit **420** may be a portion through which the blood sample S flows into the body **410**.

[0071] The movement channel **430** may be provided on the body **410** and may be configured to move the blood sample S introduced through the injection unit **420**. Here, the movement channel **430** may be referred to as a 'microchannel'. After the light (e.g., laser light with λ =approximately 532 nm) emitted through the light irradiation module **100** passes through the movement channel **430**, an ROI may be captured by the first lens **220**. Through this, an 'original speckle image' is acquired, and the original speckle image is constructed to generate random bits by the random number generation module **300**, which will be described later.

[0072] The movement channel **430** may be designed with precise dimensions of approximately 45 mm in length, approximately 45 μ m in height, and approximately 1 mm in width by applying soft photolithography to construct the microchannel. As an example, the Applicant fabricated a PDMS slab according to several standard processes, and the slab composed of PDMS Sylgard 184 A/B (Dow Corning, Korea) was bonded to the cover glass through oxygen plasma treatment.

[0073] The movement channel **430** may be composed of a vacuum generator integrated with a pump, a solenoid valve, three-way valve, and a 50 ml syringe to account for dead volume. The pump may control sample collection with a constant volume of approximately 200 μL in variable recovery mode using a solenoid valve that regulates the flow.

[0074] The discharge unit **440** may be a portion that discharges the blood sample S flowing within the movement channel **430** to the outside of the body **410**. A pumping module **500** may include a pump **510**, a movement pipe **520**, a first valve **530**, a second valve **540**, and a dead volume chamber **550**.

[0075] The pumping module **500** may be configured to be connected to the sample loading module **400** and move the blood sample S toward the sample loading module **400** through vacuum pressure.

[0076] The pump **510** is a vacuum pump, which may move the blood sample S toward the body **410** of the sample loading module **400** by creating vacuum pressure. As an example, the pump **510** may be a syringe pump. The movement pipe **520** may be a pipe through which the blood sample S is moved toward the body **410** by the vacuum pressure created by the pump **510**.

[0077] The first valve **530** may be provided in the movement pipe **520** and configured to adjust the vacuum pressure applied to the body **410**. As an example, the first valve **530** may be a solenoid valve. The second valve **540** and the dead volume chamber **550** may be provided in the movement pipe **520** to buffer the movement of the blood sample S. In this case, the dead volume chamber **550** may be coupled to the movement pipe **520** through the second valve **540**. As an example, the second valve **540** may be a 3-way valve, and the dead volume chamber **550** may be a container formed of an empty space.

[0078] The random number generation module **300** is configured to generate random numbers by processing the original speckle image generated from the image generation module **200**. As an example, the random number generation module **300** may be implemented in the form of a CPU, GPU, and AP having an operation processing function, or a combination thereof, and may be provided together with DRAM, flash memory, SSD, and various other types of memory as needed.

[0079] The random number generation module **300** includes a binary image generation unit **310**, a von Neumann extraction unit **320**, and an image scrambling unit **330**.

[0080] FIG. 4 is a conceptual diagram schematically illustrating a step of generating random numbers using a system according to the embodiment of the present invention and FIG. 5 is a block diagram schematically illustrating a processing process in the von Neumann extraction unit of the system according to the embodiment of the present invention.

[0081] The binary image generation unit **310** is configured to generate a binary image by performing a binarization process of classifying pixel values constituting the original speckle image into bit of 0 or 1. That is, the original speckle image is converted into the binary image.

[0082] The image scrambling unit **330** is configured to perform scrambling on the binary image. Scrambling code injection (SCI) refers to injecting a scrambling code into the image. The bit generation speed can be improved by scrambling the image before the image is passed through the von Neumann extractor **320**.

[0083] Both the first and second processed images described later may be images generated with scrambling performed by the image scrambling unit **330**.

[0084] The von Neumann extractor **320** generates a first processed image by applying the von Neumann post-processing method to the binary image acquired through the binary image generation unit **310**. After that, the second processed image can be generated by further applying the two-pass tuple-output von Neumann (2P-TO-VN) method to the first processed image.

[0085] Here, the von Neumann post-processing method is in contrast to the two-pass tuple-output von Neumann (2P-TO-VN) method, which may be understood as the one-pass tuple-output von Neumann (1P-TO-VN) method. That is, the 2P-TO-VN method includes a step of going through an additional pass, in addition to the 1P-TO-VN method.

[0086] FIG. 6 is an example for describing the process in FIG. 5, and the method of the present invention is not limited to the method illustrated in FIG. 6.

[0087] In the von Neumann extraction unit **320**, the binary image can be processed in the 2P-TO-VN method by first going through the 1P-TO-VN method and then going through a separate pass again.

[0088] Referring to FIG. 6, the binary image goes through a step of separating two bits into one set. Here, if one set is a duplication of the same bits, the set can be removed. As an example, the set of '00' or '11' can be deleted.

[0089] Then, in a state where one set is a duplication of the same bits and the set is removed, when a set is composed of different bits based on the remaining bits, the von Neumann extraction unit is configured to maintain only a first bit or second bit constituting the set among the first bit and second bit. In this way, the 1P-TO-VN method is completed. As an example, in the case of '01' or '10', only the first bit, such as '0' of '01' or '1' of '10', can be maintained.

[0090] After that, the von Neumann extraction unit **320** is configured to perform an additional pass. Through this, the bias of the bit can be lowered.

[0091] The 2P-TO-VN method can be understood as regrouping the set removed through the 1P-TO-VN method into a quad, which is composed of a first set and a second set, and further going through a step of determining whether to maintain removal of the first and second sets based on whether the first and second sets are the same. Here, if the first and second sets are different from each other, the first and second sets can be restored. As an example, discarded first and second sets, each of which consists of different bits, within each quad, such as '0011' or '1100', can be set to be saved again.

[0092] The classic von Neumann (CVN) debiasing method, which is known as the 1P-TO-VN method, significantly reduces the number of bits due to strict compression, while the 2P-TO-VN method provides an appropriate balance between bit retention and debiasing. The 2P-TO-VN method reevaluates the initially discarded bits to thereby generate a data volume that is more suitable for real applications, as can be seen in FIG. 6, and thus, may be an excellent alternative to the CVN described above for generating strong and usable cryptographic keys (random numbers).

[0093] FIG. 7 is a flowchart of a method according to an embodiment of the present invention, and FIG. 8 is a flowchart dividing a step **S40** of FIG. 7 in detail.

[0094] The method will be described with reference to FIGS. 7 and 8, and descriptions that overlap with the above description will be omitted.

[0095] The method according to an embodiment of the present invention includes steps **S10** to **S40**.

[0096] Step **S10** is a step of supplying a preset blood sample through the sample loading module **400**.

[0097] Step **S20** is a step of irradiating the blood sample provided by the sample loading module **400** with light through the light irradiation module **100**, and irradiating the blood sample moved to a pre-designated region of interest with light.

[0098] Step **S30** is a step of acquiring an original speckle image by capturing an image of the blood sample located in the region of interest in a preset method, using the light irradiation module **100**.

[0099] Step **S40** is a step of generating random numbers by processing the original speckle image generated in the step **S30** in a preset method.

[0100] Here, step **S40** may further include steps **S41** to **S43**.

[0101] Step **S41** is a step of generating a binary image by performing a binarization process of classifying the pixel values constituting the original speckle image into a bit of 0 or 1.

[0102] Step **S411** is a step of generating a pre-processed image by performing image scrambling on the binary image generated in the step **S41**.

[0103] Step **S42** is a step of generating a first processed image by applying the von Neumann post-processing method to the binary image acquired in the step **S41**.

[0104] Step **S43** is a step of generating a second processed image by further applying the 2P-TO-

VN method to the first processed image generated in the step S42.

[0105] FIG. 9 is a graph obtained by comparing the speckle decorrelation times of the original speckle image (OSI), the first processed image (CVN method), and the second processed image (2P-TO-VN method). The Applicant has performed validation of the results of the method of generating random numbers based on blood flow.

[0106] Referring to FIG. 9, the decorrelation time is the main measure of estimated randomness. It shows the autocorrelation curves for approximately 56 original speckle images compared to the real and scrambled random bits. When comparing the original speckle image with the derived random bits, it shows a large difference in the decorrelation time. The average decorrelation time for the original speckle image was approximately 3.05 seconds with a standard deviation of approximately +0.43, and a post-processed series of random bits of the classic von

[0107] Neumann (CVN) image, the CVN with scrambling, the 2P-TO-VN method, and the 2P-TO-VN method with scrambling images show remarkably consistent mean and standard deviation of approximately 1.542 ± 0.019 , approximately 1.538 ± 0.018 , approximately 1.551 ± 0.017 , and approximately 1.529 ± 0.013 , respectively. These closely clustered values indicate that there is low or no correlation between approximately 56 different images, which means high randomness.

[0108] FIG. 10 is a graph obtained by comparing the random number matrix characteristics of the original speckle image (OSI), the first processed image (CVN method), and the second processed image (2P-TO-VN method).

[0109] Bit uniformity and correlation difference comparison will be described with reference to FIG. 10. To address potential bias in the data, bit uniformity of the original speckle image with an average of approximately 0.626 was examined by reflecting distortion. After applying the 2P-TO-VN method, an exemplary bit uniformity of approximately 0.500 was checked for both data sets of a case where data is processed only with the 2P-TO-VN method and a case of combined with scrambling code insertion. A numerical value of approximately 0.500 may be understood as illustrating an ideal balance of '1' and '0' bits.

[0110] As illustrated in (b) to (d), further investigation using correlation analysis on approximately 56 original images and the processed speckle images shows that the processed dataset achieves a correlation value that is approximately 2,270 times much lower than the original images.

[0111] In addition, when comparing the two processed datasets, the 2P-TO-VN method including SCI processing was found to decrease by approximately 18.34% in correlation value compared to the image to which only the 2P-TO-VN method was applied, which can be understood as SCI processing further weakening the correlation.

[0112] In (c) and (d), it is shown that the correlation decreases as the correlation value outside the diagonal area approaches approximately 0.

[0113] The specific correlation metrics for the original speckle images are found to have a standard deviation of approximately 0.016264 and a mean of approximately 0.01896, and the processed data shows significant improvement in that the 2P-TO-VN images and 2P-TO-VN images with SCI, that respectively have correlation values of approximately 0.000453 and approximately 0.000343, produce much lower correlations of approximately 0.000383 and approximately 0.000296, respectively. These results can strongly emphasize the effectiveness of the 2P-TO-VN method, which achieves bit uniformity and minimizes correlation, and verify the statistical reliability of random number generation based on blood flow speckles.

[0114] FIG. 11 illustrates results of speckle bit generation rates of the first processed image (CVN method) and the second processed image (2P-TO-VN method) and the respective images.

[0115] Referring to FIG. 11, (a) is a bit generation rate graph generated from the original speckle bits. The speckle bits may consist of approximately 8.1 million bits. The means and standard deviations of the respective images are calculated to be approximately 16.65 ± 2.27 , approximately 23.37 ± 0.49 , approximately 52.35 ∓ 3.34 , and approximately 67.70 ± 2.44 , respectively.

[0116] These have been tested for unpredictability using the NIST randomness test. As for the

NIST randomness test, the Applicant performed a series of statistical tests using National Institute of Standards and Technology Statistical Test Suite (NIST) in order to determine randomness quality of the random bits.

[0117] For reference, NIST tests are designed to quantitatively evaluate the randomness of binary sequences, and the NIST test suite consists of approximately 15 individual tests, each of which is designed to quantitatively measure various aspects of the randomness of the binary sequences.

[0118] The tests include frequency, block frequency, execution, longest running (LRO), continuation, approximate entropy, and cumulative sums (Cusums) evaluations, and the evaluation performed by the Applicant involved aggregating binary sequences from approximately 56 different random bits to ensure adequate stream length for approximately 7 statistical tests.

[0119] Here, $P(s)$ is a path length distribution of a sample, s is a path length, t is a delay time, 1^* is a transmission mean free path, and τ is a characteristic decay time of a medium.

[0120] Autocorrelation will be a value between approximately 0 and approximately 1, and as the time lag increases, the corresponding value should fall closer to approximately 0, which means that there is no longer any correlation compared to the first image. In order to check bio-inspired true random beat generation, the correlation time between two consecutive images in a time series of speckled pattern images was measured at approximately 50% point of as blood passed through the movement channel **430**.

[00001] $g_1(\tau) = \int_0^\infty P(s) \exp\left[-\left(\frac{2s}{\tau}\right)\right] ds$ [Equation1]

[0121] In addition, in order to evaluate data analysis (bit uniformity, average correlation) and the performance of random bits, digitized keys are examined. Here, the bit uniformity is evaluated by how balanced the distribution (ratio) of '0' and '1' bits is within the random bits.

[00002] $\text{Bituniformity} = \frac{1}{s} \cdot \text{Math}.\sum_{i=1}^s K_i$ [Equation2]

[0122] The uniformity of the random bit sequence is often evaluated by determining the Hamming weight, which is the number of '1' bits in a binary string of length 's'. For a particular set of bits, the desired uniformity is achieved when the Hamming weight approaches approximately 0.5, reflecting an even distribution of '0' and '1' bits.

[0123] In addition, the correlation between images is investigated using a correlation matrix. In order to ensure objectivity of comparison, the random bits obtained through the method of the present invention were quantified and matched to create a uniform bit matrix to be examined (e.g., original speckle image: approximately 7.8 Mbits, CVN image: approximately 1.02 Mbits, CVN with SCI: approximately 1.8 Mbits, 2P-TO-VN image: approximately 3.3 Mbits, 2P-TO-VN with SCI: approximately 4.8 Mbits).

[00003] $r = \frac{\text{Math}.\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\text{Math}.\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\text{Math}.\sum_{i=1}^n (y_i - \bar{y})^2}}$ [Equation3]

[0124] Here, the Pearson correlation coefficient formula may be used to quantify the correlation. In Equation 3, $\bar{x}$ and $\bar{y}$ represent averages of the x values and y values, respectively, and the Pearson correlation coefficient, which is denoted by r , is closer to 0, indicating that there is no significant correlation between the two sets of random bits.

[0125] FIG. 13 is a table illustrating the results of the NIST statistical randomness test for the random numbers generated using the system according to an embodiment of the present invention, and the present Applicant can confirm that all tests passed.

[0126] The (a) of FIG. 12 is an exemplary image illustrating the original speckle image (OSI), the first processed image (CVN method), and the second processed image (2P-TO-VN method), and (b) of FIG. 12 illustrates an exemplary image to which image scrambling has been applied.

[0127] Referring to FIG. 12, the speckle pattern generated unpredictable random bits consisting of 0 and 1 bits, and additional randomness was incorporated using the image scrambling and the 2P-TO-VN method in order to improve high output and performance.

[0128] Referring to (b), the total pixels of the images containing the SCI are approximately 49980

(60×833) pixels (see (a)) and approximately 56520 (60×942) pixels (see (b)), respectively, and thus it can be seen that they are affected by scrambling.

[0129] According to an embodiment of the present invention, an innovative system for generating speckle patterns using a blood flow is presented.

[0130] In particular, the unique characteristics of blood, which is rich in various cellular components, can increase the randomness of the generated random numbers by improving the complexity and entropy of the generated speckle patterns.

[0131] The present invention can capture speckle patterns at the high speed of the system and requires only a very small amount of blood, which can improve system accessibility by minimizing both the cost and acquisition time associated with random number generation.

[0132] In addition, the system according to an embodiment of the present invention can increase the number of generated bits while providing overall unpredictability and strong security of the system, by applying the two-pass tuple-output von Neumann (2P-TO-VN) method, which is designed to prevent the generated random bits from being biased.

[0133] In addition, various other additional effects can be achieved by various embodiments of the present invention. These various effects of the present invention will be described in detail in each embodiment, or descriptions of effects that can be easily understood by those skilled in the art will be omitted.

[0134] As above, although the present invention has been described with limited embodiments and drawings, the present invention is not limited thereto, and various modifications and variations may be made by those skilled in the art in the technical field to which the present invention belongs within the technical idea of the present invention and the scope of equivalency of the claims described below.

Claims

1. A system for generating random numbers based on blood samples, comprising: a sample loading module configured to supply a preset blood sample; a light irradiation module configured to irradiate the blood sample provided by the sample loading module with light, the blood sample moved to a pre-designated region of interest (ROI) being irradiated with light; an image generation module configured to acquire an original speckle image by capturing an image of the blood sample located in the region of interest in a preset method using the light irradiation module; and a random number generation module configured to generate random numbers by processing the original speckle image generated from the image generation module in a preset method.
2. The system of claim 1, wherein the random number generation module includes a binary image generation unit configured to generate a binary image by performing a binarization process of classifying pixel values constituting the original speckle image into bit of 0 or 1.
3. The system of claim 2, wherein the random number generation module further includes a von Neumann extraction unit configured to generate a first processed image by applying a von Neumann post-processing method to the binary image acquired through the binary image generation unit.
4. The system of claim 3, wherein the von Neumann extraction unit is configured to generate a second processed image by further applying a two-pass tuple-output von Neumann (2P-TO-VN) method to the first processed image.
5. The system of claim 4, wherein the von Neumann extraction unit is configured to apply a one-pass tuple-output von Neumann (1P-TO-VN) method and process two bits as one set in a preset method, and the preset method is a method of removing one set if the one set is a duplication of the same bits.
6. The system of claim 5, wherein the von Neumann extraction unit is configured to, when a set is composed of different bits based on the remaining bits in a state where one set is the duplication of

the same bits and the set is removed, maintain only a first bit or second bit constituting the set among the first bit and second bit.

7. The system of claim 6, wherein the von Neumann extraction unit is configured to regroup the removed set into a quad through the 1P-TO-VN method, and the quad is composed of a first set and a second set, and determine whether to maintain the removal of the first and second sets based on whether the first and second sets are the same.

8. The system of claim 7, wherein the von Neumann extraction unit is configured to restore the first and second sets when the first and second sets are different from each other.

9. The system of claim 4, wherein the random number generation module is configured to further include an image scrambling unit configured to generate a pre-processed image by performing image scrambling on the binary image generated through the binary image generation unit, and the von Neumann extraction unit is configured to generate the first and second processed images for the pre-processed image generated by the image scrambling unit.

10. A method of generating random numbers based on blood samples, comprising: (a1) supplying a blood sample through a sample loading module; (a2) irradiating the blood sample provided by the sample loading module with light through a light irradiation module, the blood sample moved to a pre-designated region of interest (ROI) being irradiated with light; (a3) acquiring an original speckle image by capturing an image of the blood sample located in the region of interest in a preset method using the light irradiation module; and (a4) generating random numbers by processing the original speckle image generated in the (a3) in a preset method.

11. The method of claim 10, wherein the (a4) includes (a41) generating a binary image by performing a binarization process of classifying pixel values constituting the original speckle image into bit of 0 or 1, (a42) generating a first processed image by applying a von Neumann post-processing method to the binary image acquired in the (a41), and (a43) generating a second processed image by further applying a 2P-TO-VN scheme to the first processed image generated in the (a42).

12. The method of claim 11, further comprising: before the (a42), (a42-0) generating a pre-processed image by performing image scrambling on the binary image generated in the (a41).
